Abenezer Bekele

JUNIOR DATA ENGINEER

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PROFESSIONAL SUMMARY

Detail-oriented Computer Science graduate and Junior Data Engineer with 1+ years of experience (internship & academic) building robust, scalable data pipelines. Expert in Python and SQL for ETL/ELT development, data modeling, and workflow automation. Proven ability to translate complex datasets into actionable insights for mission-driven organizations. Proficient in applying Generative AI to optimize development workflows and building RESTful APIs for seamless data movement. Committed to data quality through rigorous unit testing and comprehensive logging.

EDUCATION

University of Wisconsin – River Falls | River Falls, WI

- Bachelor of Science in Computer Science and Information Systems | December 2024
  Relevant Coursework: Database Systems, Data Science, Software Engineering, Object-Oriented Programming, Data Structure Algorithms

CERTIFICATIONS

- IBM data science professional certificate
- Google Data Analytics Professional Certificate
- AWS Cloud Practitioner Essentials (In Progress)

WORK EXPERIENCE

Development Intern | A Better Road Foundation

- Designed, developed, and deployed end-to-end web platform; analyzed datasets to support data-informed email and web engagement strategies.
- Built custom interactive elements using JavaScript (Velo), ensuring data integrity and responsive user experience across platforms.
- Integrated Google Analytics and implemented SEO best practices, providing leadership with data-driven insights on user engagement KPIs.
- Acted as sole technical liaison between stakeholders and IS team, gathering requirements and translating organizational goals into functional solutions.

TECHNICAL SKILLS

- **Languages:** Python (Pandas, NumPy, Scikit-learn), SQL (PostgreSQL, MS SQL, MySQL), JavaScript, TypeScript.
- **Data Engineering:** ETL/ELT Pipeline Design, Data Modeling (Star/Snowflake), Data Warehousing, Automation.
- **DevOps & Tools:** Git/GitHub (Source Control), Docker (Containerization), CI/CD, Jira, Agile/Scrum.
- **Cloud & APIs:** AWS (S3, Lambda, EC2), RESTful API Development, Google Gemini API (GenAI integration).
- **Data Quality:** Unit Testing (PyTest), Logging/Monitoring, Data Wrangling, Metadata Management.

PROJECTS

FinTracker: AI-Powered Financial API | GitHub

- **API Development:** Built a secure RESTful API using Node.js and PostgreSQL to facilitate data movement across systems.
- **GenAI Integration:** Leveraged Generative AI (Google Gemini Pro) to automate the extraction of metadata from unstructured images, optimizing the data ingestion workflow.
- **Containerization:** Utilized Docker for consistent deployment environments, ensuring scalability and security.

Automated ETL & ML Pipeline: Titanic Survival & Drug Classification | GitHub

- **Data Quality & Wrangling:** Built a robust ETL pipeline for the Titanic and Drug-Classification datasets, handling missing values, feature scaling, and normalization to ensure model-ready data.
- **Predictive Analytics:** Developed and deployed machine learning models (Random Forest, Decision Trees) to predict outcomes, achieving high accuracy through hyperparameter tuning.
- **Unit Testing:** Applied source control (Git) and modular coding practices to ensure the pipeline was reusable and maintainable.

Public Interest Data Pipeline: Chicago Urban Analysis | GitHub

- **Complex Dataset Handling:** Engineered a data pipeline to process diverse urban datasets (socio-economic, education, and population) to identify service gaps in metropolitan areas.
- **Analysis & Modeling:** Utilized Python and Jupyter to clean and normalize "messy" public data, creating a clean core system for secondary analysis.
- **Visualization:** Developed automated reporting modules to communicate technical demographic trends to non-technical stakeholders.

Relational Database Design & SQL Optimization (COVID-19) | GitHub

- **Advanced SQL:** Authored 12+ complex SQL queries using Window Functions, CTEs, and Self-Joins to transform raw global health data into structured analytical views.
- **Performance Tuning:** Optimized query execution by implementing strategic indexing and stored procedures, improving data retrieval speeds for regional summary reporting.
- **Data Governance:** Applied NULL filtering and data validation constraints to maintain high data integrity standards across all interfaces.